Amazing Game II

Amy and Bruno are playing another game in a maze! The maze has N rooms (numbered from 1 to N) and M bidirectional passageways, where each passageway connects two different rooms U_i and V_i. It is possible to travel between any two rooms in the maze using a sequence of passageways. Amy has chosen a path with K rooms: $A_1, A_2, \ldots, A_K$, and you are guaranteed that there is a passageway between consecutive rooms in her path. Note that the same room could appear multiple times in Amy's path.

Amy and Bruno will play their game Q times, where $Q < K$. In the ith game, Amy starts in room A_i (the ith room in her path) and Bruno starts in room B_i. They then play a game with $K - i$ rounds. In each round:

- Amy walks from her current room to the next room in her path. That is, if she is currently in room A_j, then she walks to room A_{j+1}.
- Bruno can choose to either stay in his current room or walk to any room which is connected via a single passageway to his current room.

In each round, Bruno earns a point if he stays still and Amy moves into his current room. The game ends when Amy reaches the end of her path.

Bruno has come to you for help: in each of the Q games, what is the maximum number of points that he can earn?

Partial points are available for just determining whether Bruno can earn any points. See the scoring section for details.

Subtasks and Constraints

For all subtasks:

- $2 \leq N \leq 1000$.
- $N - 1 \leq M \leq 2000$.
- $3 \leq K \leq 200\,000$.
- $1 \leq Q < K$.
- $1 \leq U_i < V_i \leq N$ for all i.
- All passageways are different. That is $(U_i, V_i) \neq (U_j, V_j)$ for all $i \neq j$.
- It is possible to travel between any two rooms using a sequence of passageways.
- $1 \leq A_i \leq N$ for all i.
- There is a passageway between rooms A_i and A_{i+1} for all $1 \leq i \leq K - 1$.
- $1 \leq B_i \leq N$ for all i.

Additional constraints for each subtask are given below.

Subtask	Points	Additional constraints
1	10	$Q = 1$, $K = 3$, and $A_1 = B_1$.
2	30	$Q = 1$, $K \leq 1000$, and $A_1 = B_1$.
3	25	$Q = 1$ and $A_1 = B_1$.
4	15	$A_i = B_i$ for all i.
5	20	No additional constraints.

Input

- The first line of input contains the integers N and M.
- M lines follow, describing the passageways. The ith line contains the two integers U_i and V_i.
- The next line contains the integer K.
- The next line contains the K integers $A_1, A_2, \ldots, A_K$.
- The next line contains the integer Q.
- The next line contains the Q integers $B_1, B_2, \ldots, B_Q$.

Output

Output Q lines, where the ith line contains the maximum number of points that Bruno can earn in the ith game.

Scoring

You will receive 30% for just determining whether the answer is 0 or non-zero in every game. In particular:

- If your output is correct, you will receive 100% for that test case. Otherwise,
- You will receive 30% for a test case if:
  - You correctly answer every game where the maximum number of points Bruno can earn is 0, and
  - You output a non-zero integer for any games where Bruno can earn at least 1 point.

  Otherwise,
- You will receive 0% for that test case.

Your score for a subtask will be the **minimum** score of all test cases in the subtask (multiplied by the number of points you can score in the subtask).

Sample Input 1

```
5 6
1 2
1 3
2 3
3 4
3 5
4 5
3
4 3 2
1
4
```

Sample Output 1

```
0
```

Sample Input 2

```
5 6
1 2
1 3
2 3
3 4
3 5
4 5
3
1 2 1
1
1
```

Sample Output 2

```
1
```

Sample Input 3

```
5 6
1 2
1 3
2 3
3 4
3 5
4 5
7
1 2 3 4 5 3 5
5
1 3 2 2 2
```

Sample Output 3

```
3
3
1
1
0
```

Explanation

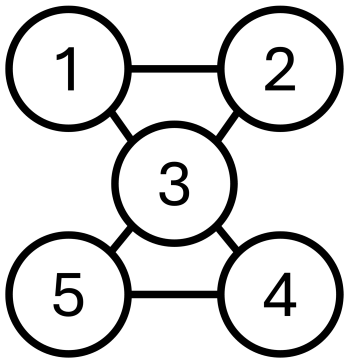


Figure 1: All three sample cases involve the same maze, shown above

In the first sample case, Amy's path is $4 \to 3 \to 2$. There is $Q = 1$ game, where Bruno and Amy both start in room 4. It is impossible for Bruno to earn any points.

In the second sample case, Amy's path is $1 \to 2 \to 1$. There is $Q = 1$ game, where Amy and Bruno both start in room 1. Bruno can earn one point as follows:

- Amy walks from room $A_1 = 1$ to $A_2 = 2$, and Bruno remains in room 1.
- Amy walks from room $A_2 = 2$ to $A_3 = 1$, and Bruno remains in room 1. Bruno earns a point.

In the third sample case, Amy's path is $1 \to 2 \to 3 \to 4 \to 5 \to 3 \to 5$. There are $Q = 3$ games. In the first game, Amy and Bruno both start in room 1 and Bruno can earn three points as follows:

- Amy walks from room $A_1 = 1$ to $A_2 = 2$, and Bruno walks from room 1 to room 3.
- Amy walks from room $A_2 = 2$ to $A_3 = 3$, and Bruno remains in room 3. Bruno earns a point.
- Amy walks from room $A_3 = 3$ to $A_4 = 4$, and Bruno walks from room 3 to room 5.
- Amy walks from room $A_4 = 4$ to $A_5 = 5$, and Bruno remains in room 5. Bruno earns a point.
- Amy walks from room $A_5 = 5$ to $A_6 = 3$, and Bruno remains in room 5.
- Amy walks from room $A_6 = 3$ to $A_7 = 5$, and Bruno remains in room 5. Bruno earns a point.

In the second game, Amy starts in room 2 and Bruno starts in room 3. Bruno can earn three points as follows:

- Amy walks from room $A_2 = 2$ to $A_3 = 3$, and Bruno remains in room 3. Bruno earns a point.
- Amy walks from room $A_3 = 3$ to $A_4 = 4$, and Bruno walks from room 3 to room 5.
- Amy walks from room $A_4 = 4$ to $A_5 = 5$, and Bruno remains in room 5. Bruno earns a point.
- Amy walks from room $A_5 = 5$ to $A_6 = 3$, and Bruno remains in room 5.
- Amy walks from room $A_6 = 3$ to $A_7 = 5$, and Bruno remains in room 5. Bruno earns a point.

In the third game, Amy starts in room 3 and Bruno starts in room 2. Bruno can earn one point as follows:

- Amy walks from room $A_3 = 3$ to $A_4 = 4$, and Bruno walks from room 2 to room 3.
- Amy walks from room $A_4 = 4$ to $A_5 = 5$, and Bruno remains in room 3.
- Amy walks from room $A_5 = 5$ to $A_6 = 3$, and Bruno remains in room 3. Bruno earns a point.
- Amy walks from room $A_6 = 3$ to $A_7 = 5$, and Bruno remains in room 3.

In the fourth and fifth games, Bruno can earn 1 and 0 points respectively.